

Name: BODHU RISHIKESH

College: AVN Institute of Engineering and Technology

Branch: Artificial Intelligence & Machine Learning

Internship Task: Task 1 – Linear Regression House Price Predictor
House Price Prediction Using Linear Regression

Objective

The objective of this project is to build and evaluate a Linear Regression model using the California Housing Dataset. The model predicts house prices based on various housing features such as median income, house age, average rooms, population, latitude, and longitude.

Dataset Description

The California Housing Dataset contains 20,640 records and 9 attributes.

Features used:

- MedInc (Median Income)
- HouseAge
- AveRooms
- AveBedrms
- Population
- AveOccup
- Latitude
- Longitude

Target Variable:

- HousePrice

Tools and Libraries Used

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn
- Jupyter Notebook

Loading California Housing Dataset...

First 5 Rows:

	MedInc	HouseAge	AveRooms	AveBedrms	Population	AveOccup	Latitude \
0	8.3252	41.0	6.984127	1.023810	322.0	2.555556	37.88
1	8.3014	21.0	6.238137	0.971880	2401.0	2.109842	37.86
2	7.2574	52.0	8.288136	1.073446	496.0	2.802260	37.85
3	5.6431	52.0	5.817352	1.073059	558.0	2.547945	37.85
4	3.8462	52.0	6.281853	1.081081	565.0	2.181467	37.85

	Longitude	HousePrice
0	-122.23	4.526
1	-122.22	3.585
2	-122.24	3.521
3	-122.25	3.413
4	-122.25	3.422

Dataset Shape:

(20640, 9)

Dataset Information:

<class 'pandas.core.frame.DataFrame'>

RangeIndex: 20640 entries, 0 to 20639

Data columns (total 9 columns):

#	Column	Non-Null Count	Dtype
0	MedInc	20640 non-null	float64
1	HouseAge	20640 non-null	float64
2	AveRooms	20640 non-null	float64
3	AveBedrms	20640 non-null	float64
4	Population	20640 non-null	float64
5	AveOccup	20640 non-null	float64
6	Latitude	20640 non-null	float64
7	Longitude	20640 non-null	float64
8	HousePrice	20640 non-null	float64

dtypes: float64(9)

memory usage: 1.4 MB

None

Statistical Summary:

	MedInc	HouseAge	AveRooms	AveBedrms	Population	\
count	20640.000000	20640.000000	20640.000000	20640.000000	20640.000000	
mean	3.870671	28.639486	5.429000	1.096675	1425.476744	
std	1.899822	12.585558	2.474173	0.473911	1132.462122	
min	0.499900	1.000000	0.846154	0.333333	3.000000	
25%	2.563400	18.000000	4.440716	1.006079	787.000000	
50%	3.534800	29.000000	5.229129	1.048780	1166.000000	
75%	4.743250	37.000000	6.052381	1.099526	1725.000000	
max	15.000100	52.000000	141.909091	34.066667	35682.000000	

	AveOccup	Latitude	Longitude	HousePrice
count	20640.000000	20640.000000	20640.000000	20640.000000
mean	3.070655	35.631861	-119.569704	2.068558
std	10.386050	2.135952	2.003532	1.153956
min	0.692308	32.540000	-124.350000	0.149990
25%	2.429741	33.930000	-121.800000	1.196000
50%	2.818116	34.260000	-118.490000	1.797000
75%	3.282261	37.710000	-118.010000	2.647250
max	1243.333333	41.950000	-114.310000	5.000010

Missing Values:

MedInc	0
HouseAge	0
AveRooms	0
AveBedrms	0
Population	0
AveOccup	0
Latitude	0
Longitude	0
HousePrice	0

dtype: int64

Exploratory Data Analysis (EDA)

Data Inspection

- Dataset contains 20,640 rows and 9 columns.
- No missing values were found in any feature.
- All columns are numerical.

Feature Distribution

Histograms were used to visualize the distribution of all variables.

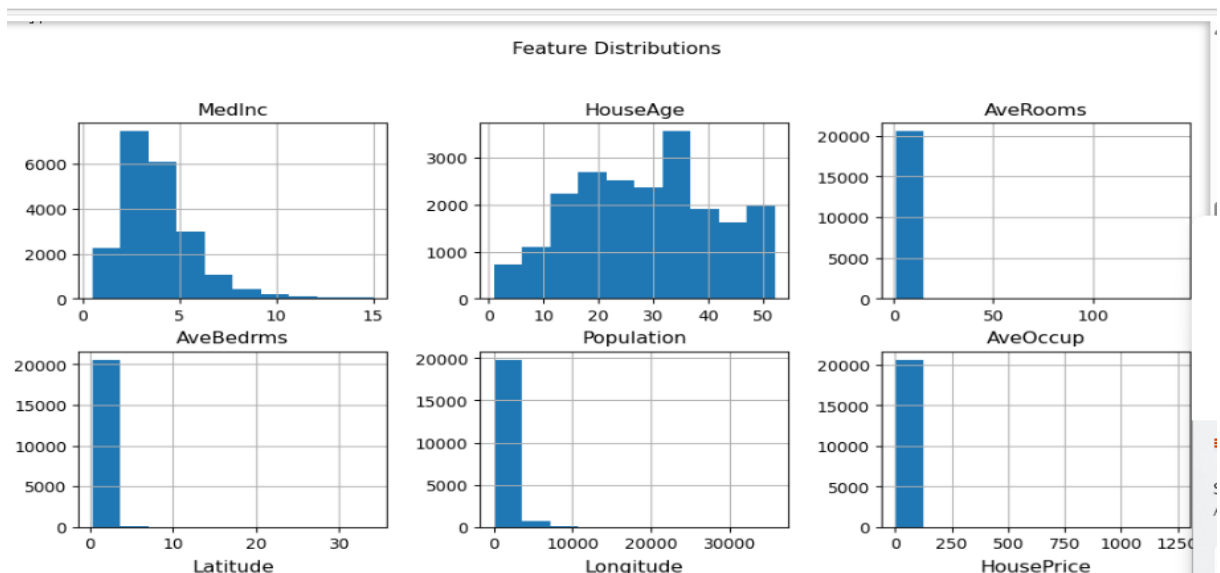
The median income feature showed a strong variation across different regions.

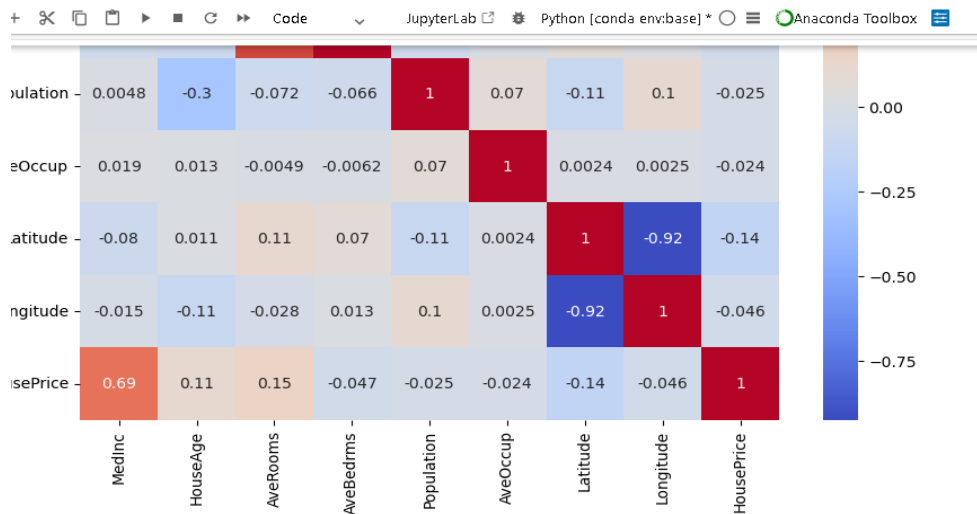
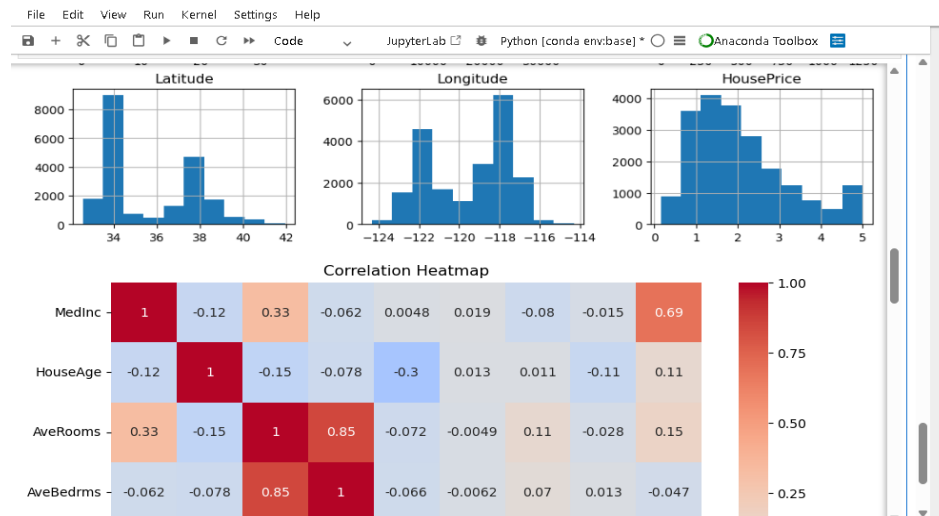
Correlation Analysis

A correlation heatmap was generated to understand relationships between variables.

Key Findings:

- Median Income (MedInc) has the strongest positive correlation with House Price (0.69).
- Latitude and Longitude show geographical influence on house prices.
- Average Rooms and Average Bedrooms are highly correlated.





The dataset was explored using histograms and a correlation heatmap. The feature distributions show variation among housing attributes. The heatmap indicates that Median Income has the strongest positive correlation with house prices. No major missing values were found in the dataset.

Model Development and Evaluation

Model Used

Linear Regression was selected because it is simple, interpretable, and effective for predicting continuous values.

Data Splitting

- Training Data: 80%
- Testing Data: 20%

Performance Metrics

<u>Metric</u>	<u>Value</u>
Mean Absolute Error (MAE)	0.533
Root Mean Squared Error (RMSE)	0.746
R ² Score	0.576

The dataset was split into 80% training data and 20% testing data using `train_test_split()`. A Linear Regression model was trained on the training dataset and evaluated on the testing dataset.

Visualization

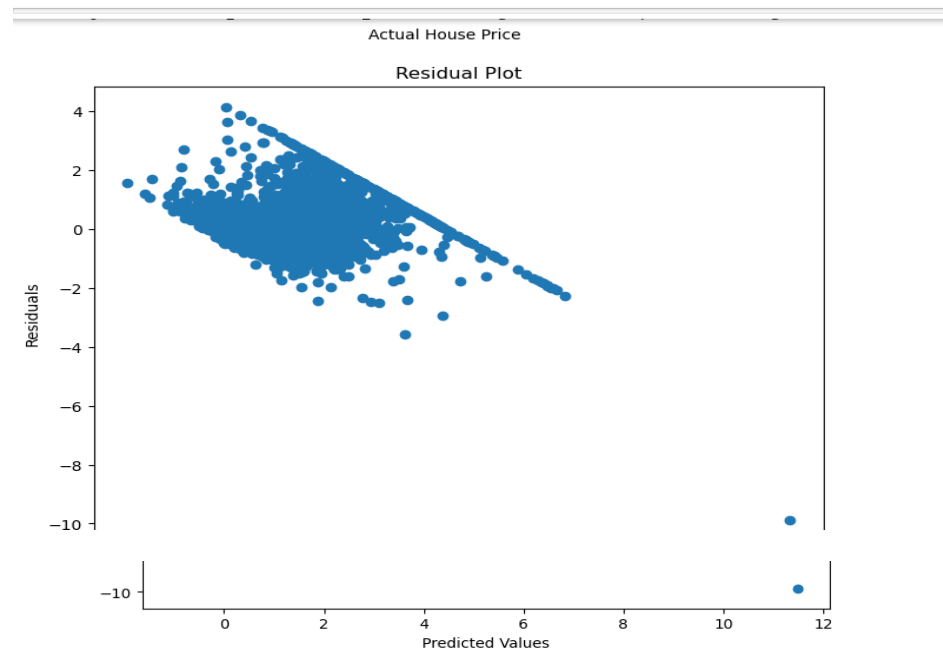
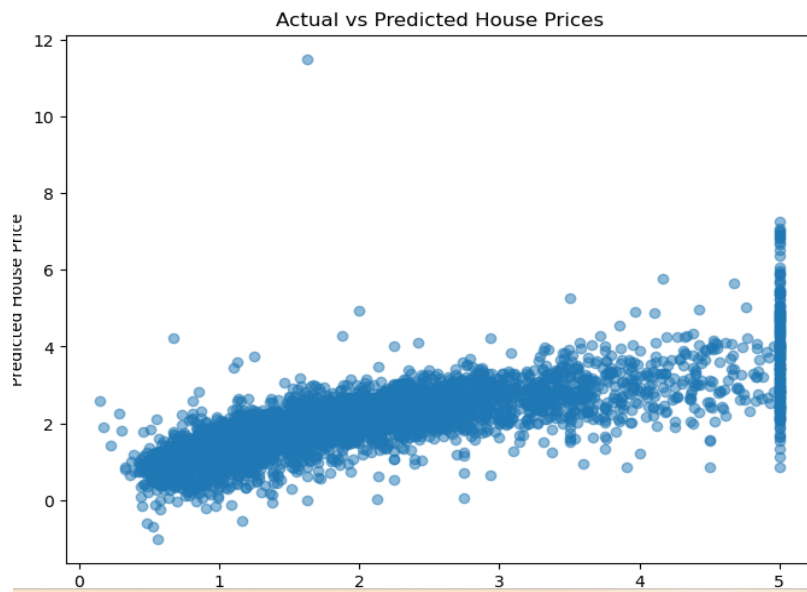
Actual vs Predicted and Residual plots were generated to evaluate model performance.

```
Training Data Shape: (16512, 8)
Testing Data Shape: (4128, 8)

Model Training Completed

===== MODEL PERFORMANCE =====
Mean Absolute Error (MAE): 0.533
Root Mean Squared Error (RMSE): 0.746
R2 Score: 0.576
```

.....



Sample Predictions:

	Actual	Predicted
0	0.47700	0.719123
1	0.45800	1.764017
2	5.00001	2.709659
3	2.18600	2.838926
4	2.78000	2.604657
5	1.58700	2.011754
6	1.98200	2.645500
7	1.57500	2.168755
8	3.40000	2.740746
9	4.46600	3.915615

Project Completed Successfully!

CONCLUSION

The Linear Regression model achieved an R^2 score of 0.576, indicating moderate predictive performance. Median Income was identified as the most influential feature affecting house prices. This project provided hands-on experience with the complete machine learning workflow including data preprocessing, exploratory data analysis, model training, evaluation, and visualization.